

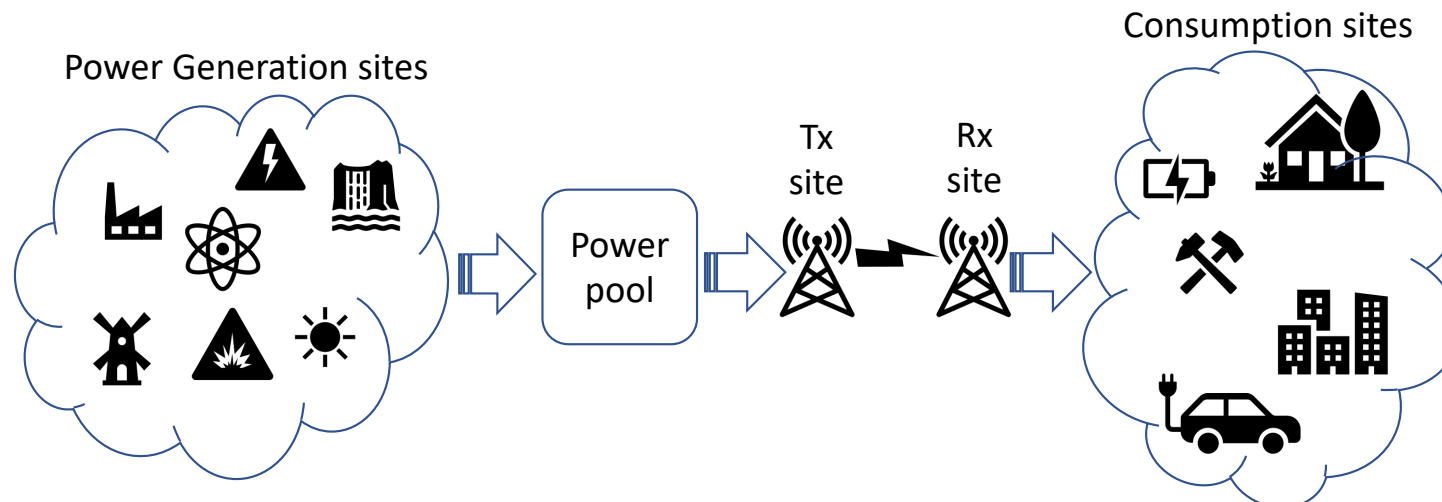
On Using NDN to Vertically Secure Smart Power Distribution

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- A smart grid is composed of numerous subsystems which continuously exchange updates and information that can be categorized as
 - technical data corresponding to energy profiles
 - management information related to billing, pricing, etc.
- NDN's data centric communication model can vertically secure the SG communication
 - Specialized naming conventions and optimal routing with the request-response approach
 - Data-centric security - secure transaction from end-to-end (power generation to consumption)

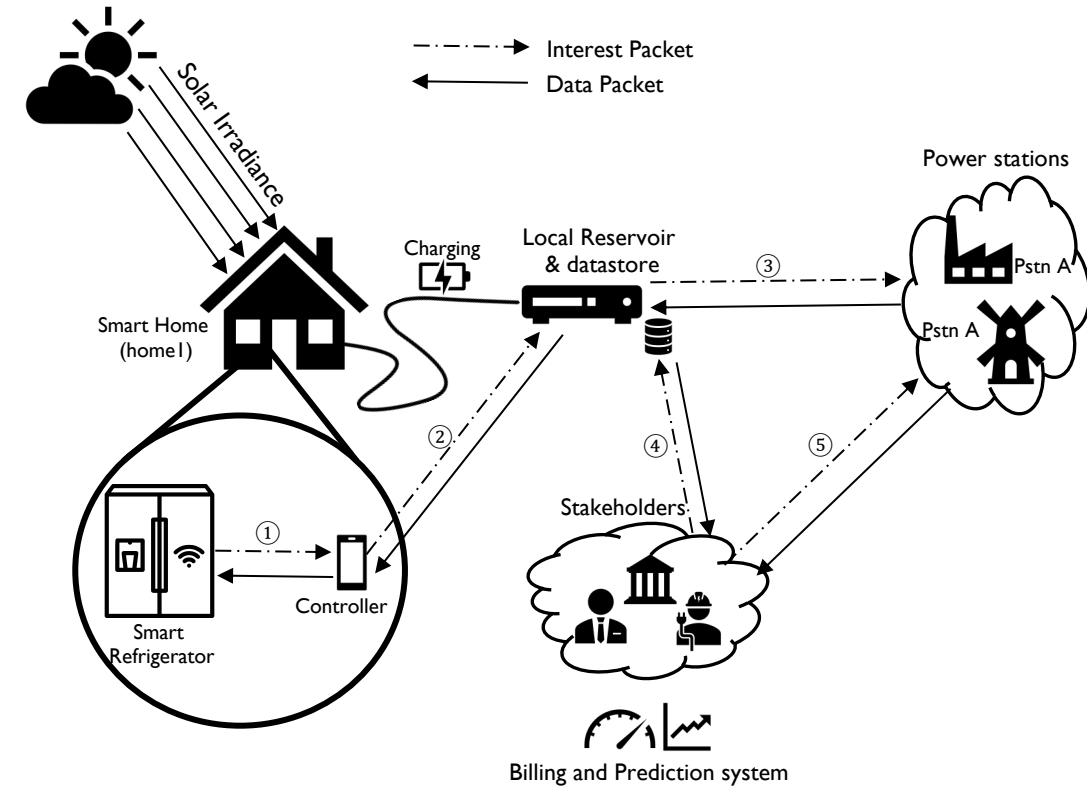


SmartGrid Communication Issues

- Common issues
 - Power systems are usually legacy systems – do not interface very well.
 - Consumers can vary from IoT devices (smart homes) to industrial machinery etc.
 - Generation sites are at remote places and power pooling involves multiple third parties
- Security concerns
 - Information and power flows through different channels and the media and is vulnerable to attacks.
 - Multiparty involvement leads to complex and difficult access control and trust policies.
- NDN based solutions
 - Optimal paths for routing information
 - Opportunistic caching at nodes (in-network storage) for frequently requested information - power consumed in retrieving information from producer is optimized.
 - Access control policies and the trust schema enables trusted and secure information exchange even in a highly distributed setup
 - Ease of information access for predictive analysis.

Example Communication

- Refrigerator requests controller for 250 watts of power
“/pndn/homeI/sg/refrigeratorI/kitchen/250W”
- Controller forwards this request to the solar power reserve
“/pndn/homeI/sg/controller/reservoir/250W”
- Solar power reservoir based on availability, either sends the requested power and an acknowledgement or further requests the grid
“/pndn/homeI/sg/pstnA /200W/IPM” for the remaining power.
- Stakeholders (user / power company), can periodically check consumption report
 - for billing and analytics
 - Proactive power distribution
 - “/pndn/homeI/sg/owner/cooling_report/021720 /1300”
- “/pndn/FPL/sg/pstnB /022020/homeI”



Smart Home connected to smart grid

- NDN-based Smart-grid operations
 - Routing and forwarding
 - Access control
 - Trust schema
 - Security challenges and solutions